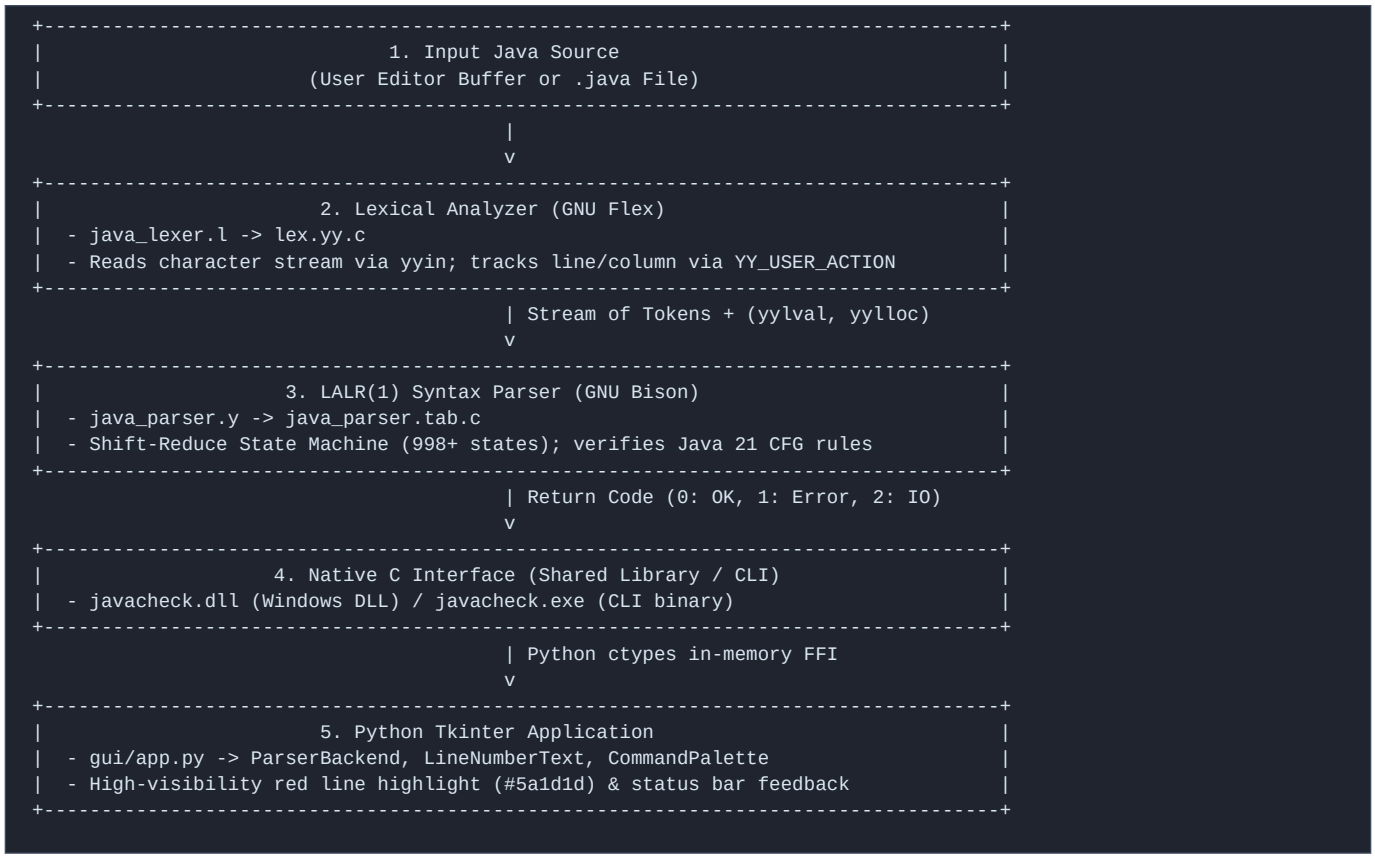
COURSE-BASED PROJECT DOCUMENTATION | DOC-05

System Architecture & Data Flow

Modular Front-End Compiler Architecture & Subsystem Boundaries

1. End-to-End System Architecture

The project is structured into four distinct, loosely-coupled layers: Lexical Analysis, Syntax Parsing, Native C Backend Interface, and the Graphical Application Layer.



2. Component Responsibilities

- Flex Scanner: Strips comments/whitespace, tracks line coordinates, emits token codes.
- Bison Parser: Executes LALR(1) shift-reduce transitions, catches errors via yyerror().
- Native C Engine: Resets state, opens file streams, formats standardized error strings.
- Python ctypes FFI: Loads javacheck.dll in memory, allocates string buffers, executes checks.
- Tkinter GUI: Renders dark editor, synchronizes line numbers, applies visual error tags.